

Alexis DHERMY

Applicant for M.S. Program in Artificial Intelligence – South Korea (Spring 2027)

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EDUCATION

- Sep 2021 – **Master of Science in Engineering (French *Diplôme d'Ingénieur*)** | EPF Engineering School, Cachan, France
Aug 2026 *Major: Intelligent Systems and Digital Technologies – AI & Cloud Computing Track*
- GPA: 15.87/20 | Rank: Top 3% of cohort
 - 2-year intensive preparatory cycle (Mathematics & Physics), 1-year general engineering, 2-year specialization
 - Thesis: LLM-Based RAG Architecture for Enterprise Systems (*Viridien*) | Advisor: Prof. Zehira Haddad
 - Relevant Coursework: Generative AI, Deep Learning, Machine Learning, Probability & Statistics, Linear Algebra
- Sep 2024 – **Exchange Semester in Artificial Intelligence** | Sungkyunkwan University (SKKU), Seoul, South Korea
Dec 2024 *College of Computing and Informatics – School of Convergence*
- GPA: 4.10/4.5
 - Relevant Coursework: Reinforcement Learning, Natural Language Processing, Large Language Models, AI Search Algorithms, Data Science

HONORS & AWARDS

- **National Engineering Entrance Merit Distinction** (*Concours Avenir – Grand Classé*): Selected in the top tier of 10,000+ candidates in one of France's most competitive post-high school engineering entrance exams; earning full exam exemption.

RESEARCH INTERESTS

- LLMs, Reasoning Models, AI Agents, Efficient Inference
- Reinforcement Learning, Embodied AI, Robotics, World Models

ACADEMIC PROJECTS

- Nov 2024 – **Music Recommendation Using NLP** ([Code & Report](#)) | Sungkyunkwan University (SKKU)
Dec 2024
- Developed an end-to-end NLP pipeline (NLTK, SpaCy, BERT) to perform tokenization, POS tagging, lemmatization, and sentiment analysis on top Spotify tracks across France, Spain, and Korea.
 - Built a cross-cultural recommendation engine that maps semantic and emotional features to recommend French/Spanish tracks based on a target Korean song.
- Nov 2024 – **Teaching a Robot to Walk with Reinforcement Learning** ([Presentation](#)) | Sungkyunkwan University (SKKU)
Dec 2024
- Trained a 2D bipedal robot in OpenAI Gym (BipedalWalker-v3) to navigate complex terrain through deep reinforcement learning.
 - Designed custom reward shaping combining speed, distance, and joint stability with a step-based delay threshold, driving a 3x increase in distance and velocity over baseline Advantage Actor-Critic (A2C).
 - Implemented Proximal Policy Optimization (PPO) to mitigate policy collapse, boosting average episode moving distance by 132%.

PROFESSIONAL EXPERIENCE

- Feb 2026 – **AI Software Engineer** (Master's Thesis Internship) | *Viridien*, Massy, France
Jul 2026 *R&D Department – Seismic Imaging*
- Conducted applied AI research to architect an enterprise advanced RAG system (LlamaIndex, Python), parsing complex geophysical tomography documentation and DSLs to boost response accuracy from 54% to 90%.
 - Optimized ingestion and retrieval via hierarchical chunking, LLM metadata extraction, hybrid search (dense + BM25), and reranking.
 - Benchmarked multi-variable configurations (chunk sizes, embedding models, fusion strategies, LLMs, and top-k) to systematically double retrieval efficiency.
 - Engineered a Model Context Protocol (FastMCP) server exposing 5 custom tools for database retrieval and agent prompt skills.
 - Containerized production infrastructure for the official enterprise AI platform using Docker, Qdrant, MongoDB, and Ollama.

- Sep 2025 – **Human-Robot Interaction Developer** (Corporate Academic Project) | *Air France Industries*, Orly, France
Jan 2026
- Deployed autonomous visitor onboarding and multimodal interaction flows on the Pepper humanoid robot for the Air France Roissy MRO Lab by engineering native Android applications (Kotlin, SoftBank QiSDK).
 - Reverse-engineered legacy robot architecture and internal systems to build a real-time engagement pipeline combining sensor-based presence detection, voice commands, and tablet UI.
 - Modernized the robot's dialogue pipeline by integrating Mistral LLM APIs, upgrading basic command handling into dynamic, context-aware conversational flows.
- Jul 2025 – **Backend & Cloud Developer** (Freelance) | *EPF Projets*, Cachan, France
Oct 2025
- Engineered the backend for "Carbon Track", a carbon-footprint mobile application, building REST APIs (Node.js, Express) connected to ADEME and Open Food Facts environmental databases.
 - Designed a hybrid database & serverless backend combining Firebase (Cloud Functions, Firestore, Auth) with local SQLite for offline data persistence.
 - Automated mobile build and release pipelines using the Expo/EAS CI/CD stack to test, package, and deploy the native application onto the Google Play Store.

TECHNICAL SKILLS

- **Programming:** Python, Java, SQL
- **ML, DL & RL:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Gymnasium
- **NLP & Gen AI:** Hugging Face, LlamaIndex, LangChain, FastMCP, Ollama, Vector Databases, Retrieval Strategies
- **Systems & Cloud:** Linux, Docker, Git, AWS, GCP, Azure
- **Languages:** English (Professional, TOEIC 895), French (Native), Korean (Basic, SKKU Coursework)